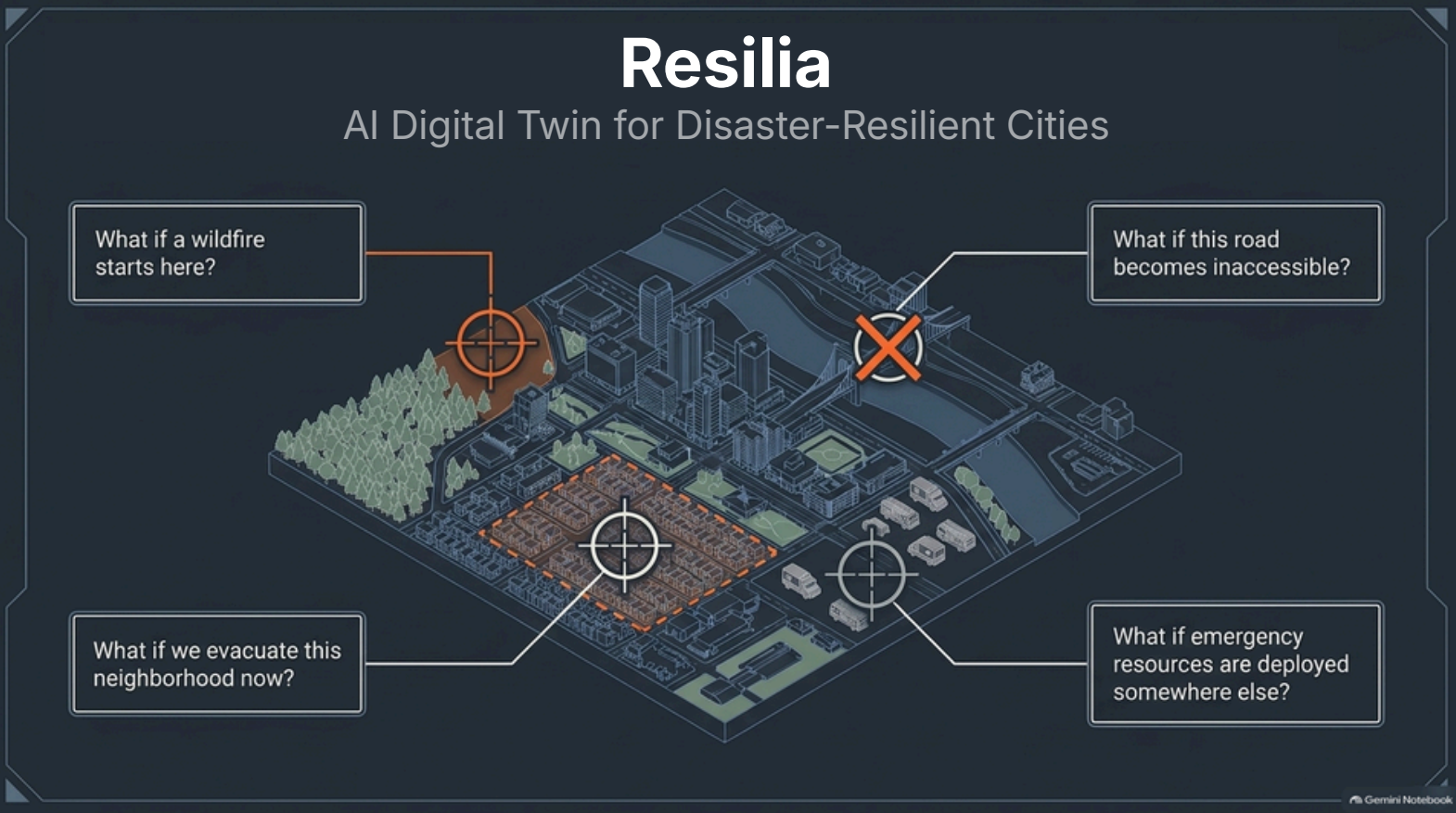
Resilia

AI Digital Twin for Disaster-Resilient Cities



"What if a city could experience a disaster before it happens?"

RESILIA is an **AI-powered Digital Twin** that transforms real cities into dynamic simulation environments. It combines geospatial, environmental, infrastructure and real-time data to model how disasters evolve and lets authorities test **"What If?" scenarios** before making decisions in the real world.

- **What if a wildfire starts here?**
- **What if this road becomes inaccessible?**
- **What if we evacuate this neighborhood now?**
- **What if emergency resources are deployed somewhere else?**

RESILIA simulates the consequences, identifies threatened buildings, infrastructure and evacuation corridors, and helps decision-makers compare interventions and choose the most resilient response.

Beyond Wildfires

Wildfire is our initial use case. The same simulation engine can model floods, post-earthquake response, hurricanes, volcanoes, and conflict-driven evacuations turning RESILIA into a scalable, multi-disaster urban resilience platform.

Innovation & Competitive Advantage

Unlike systems focused primarily on detection, alerts or static risk maps, RESILIA combines:

AI Prediction + Digital Twins + Geospatial Intelligence + What-If Simulation + Decision Support

Its architecture is designed to be:

- **Scalable** — from a neighborhood to an entire metropolitan area.
- **Disaster-agnostic** — the same platform can support multiple disaster types.
- **Scenario-driven** — decision makers can test "what-if" interventions rather than only observe current conditions.
- **AI-native** — natural-language interaction can allow non-technical users to query the digital twin and explore scenarios.
- **Cross-border ready** — the platform can be adapted to different cities, geographic environments, data sources and emergency-management structures.

Target & Business Value

RESILIA is designed for municipalities, civil protection agencies, emergency services, infrastructure operators and urban planners.

It can be delivered as a **B2G/B2B SaaS platform**, with different levels of access:

- **City / Agency Platform** — annual subscription for digital-twin infrastructure, simulation and AI decision support
- **Scenario & Risk Analysis** — paid advanced simulations and specialized studies
- **Infrastructure / Utility Intelligence** — enterprise deployments for critical infrastructure operators
- **Resident & Community Layer** — optional preparedness and personalized risk-information services

Vision

The city can test interventions before deploying them.

RESILIA gives cities a place to ask "What if?" and discover the answer before reality does.

The Team

The team consists of four university students with a variety of backgrounds, from psychologists to computer engineers, skills which allowed us to come up with and realize our idea



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